

CS757 Final Project Report

Paper:- Towards Biologically Plausible Deep Learning

Indra Anuraag Gade
G01443454

1. Introduction

The success of deep learning has largely been driven by the backpropagation algorithm, which computes gradients through the entire network. However, backpropagation has long been criticized for its lack of biological plausibility, particularly its reliance on symmetric weight transport and global error signals, which lack support from neuroscience.

In the paper *Towards Biologically Plausible Deep Learning*, Bengio et al. propose a novel learning framework that replaces backpropagation with *target propagation*—a more biologically plausible alternative. Instead of propagating errors, this approach trains local encoder–decoder pairs to propagate *targets* and uses *iterative inference* to refine hidden states. The authors show that this mechanism can be interpreted as a form of variational EM and is closely related to denoising autoencoders and generative modeling.

In this project, I reimplement the proposed model from scratch in PyTorch, faithfully reproduce the paper’s experiments on the MNIST dataset, and extend the work in two directions:

- Modifying the architecture to assess how increased capacity affects generation and inference;
- Applying the method to *Fashion-MNIST*, a more challenging image dataset.

My objective is to evaluate both the *biological plausibility* and *practical performance* of the model, and to analyze how architectural changes influence the benefits of iterative inference during generation.

2. Paper Summary

The paper *Towards Biologically Plausible Deep Learning* (Bengio et al., 2015) proposes a deep learning framework designed to better align with known biological learning mechanisms. The authors critique the standard backpropagation algorithm for its biological implausibility, such as symmetric weight transport and global error propagation, and offer an alternative based on target propagation and iterative inference.

Key Contributions

- **Target Propagation:** Instead of propagating gradients backward, the model trains local encoder–decoder pairs. Each decoder learns to reconstruct the activations of the layer below, effectively sending target activations backward layer by layer.
- **Denoising Autoencoders:** Each layer pair is trained with Gaussian noise added to inputs, using a denoising reconstruction loss. This enhances the robustness and expressiveness of learned representations and is crucial for biological plausibility.
- **Iterative Inference via Feedback:** During both training and sampling, the model performs iterative inference to refine hidden states. Feedback through the encoder–decoder mismatch adjusts internal representations toward more likely latent configurations (Algorithm 2).
- **Variational EM Interpretation:** The paper provides a probabilistic justification of the model, interpreting it as an instance of variational Expectation-Maximization (EM). The inference dynamics approximate the E-step by pushing latent variables closer to the data-driven posterior.

- **Generative Sampling & Evaluation:** To generate samples, the model draws from a Gaussian prior over the top-layer latent variables and decodes them to pixel space. Parzen window log-likelihood is used to estimate the quality of generated samples.

Experiments

The authors evaluate their method on the MNIST dataset for unsupervised generative modeling. Results show that samples generated using iterative inference are significantly more realistic than those generated via direct decoding from prior samples. This supports the central hypothesis: feedback-based inference guides the model toward latent states that resemble those encountered during training.

3. Implementation and Reproduction

To faithfully reproduce the experiments from the paper, we reimplemented the model from scratch in PyTorch, as the original codebase was written in **Theano**, a now-deprecated framework.

3.1 Model Architecture (Original)

The original model consists of:

- **Encoder:**

$$F_1(x) = \text{Softplus}(W_1x + b_1), \quad F_2(h_1) = \text{Sigmoid}(W_2h_1 + b_2)$$

- **Decoder:**

$$G_2(h_2) = \text{Softplus}(V_2h_2 + c_2), \quad G_1(h_1) = \text{Sigmoid}(V_1h_1 + c_1)$$

The forward pass encodes the input x into latent variables h_1 and h_2 , and reconstructs $\hat{x} = G_1(G_2(h_2))$. Each encoder–decoder pair is trained with a denoising objective, using Gaussian noise added to inputs and intermediate representations.

3.2 Inference Mechanism

The model uses *iterative inference*, as described in Algorithm 2 of the paper. Given input x , latent variables are refined using encoder–decoder feedback:

$$h_2 \leftarrow h_2 + \alpha (F_2(h_1) - F_2(G_2(h_2)))$$

$$h_1 \leftarrow h_1 + \alpha (F_1(x) - F_1(G_1(h_1))) + \alpha (G_2(h_2) - h_1)$$

This process pushes hidden states toward values more consistent with the decoder and closer to the data manifold.

3.3 Training Objective

The model minimizes four local reconstruction losses:

- **Loss_{F₁}:** Predicting h_1 from a noisy input
- **Loss_{G₁}:** Reconstructing x from h_1
- **Loss_{F₂}:** Predicting h_2 from a noisy h_1
- **Loss_{G₂}:** Reconstructing h_1 from h_2

These losses are summed to compute the total loss. We trained the model using RMSProp (learning rate $= 3 \times 10^{-5}$) with batch size 100 and $k = 15$ inference steps.

3.4 Reproduction Results on MNIST

We trained the model for 30 epochs on the MNIST dataset. Parzen window log-likelihood was used to evaluate sample quality. Visual inspection of generated samples and log-likelihoods confirmed effective learning and high sample fidelity.

- **Final Avg Loss (Epoch 30):** ~ 64.9

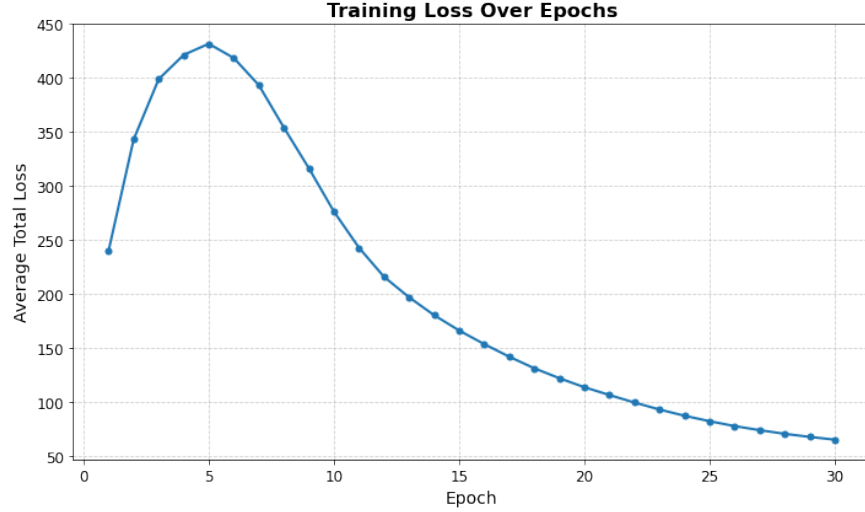


Figure 1: Training Loss over Epochs

- **Parzen Log-Likelihood (Epoch 30):**
 - With Inference: 209.19
 - Without Inference: 148.82

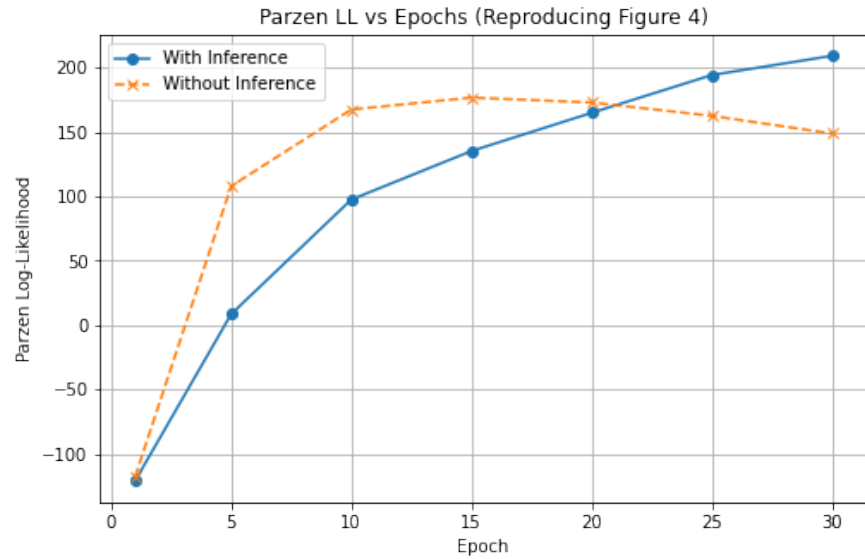


Figure 2: Parzen LL vs Epochs

These results are in line with trends reported in the paper (e.g., $LL \approx 236$ with inference), confirming the effectiveness of the reimplementation.

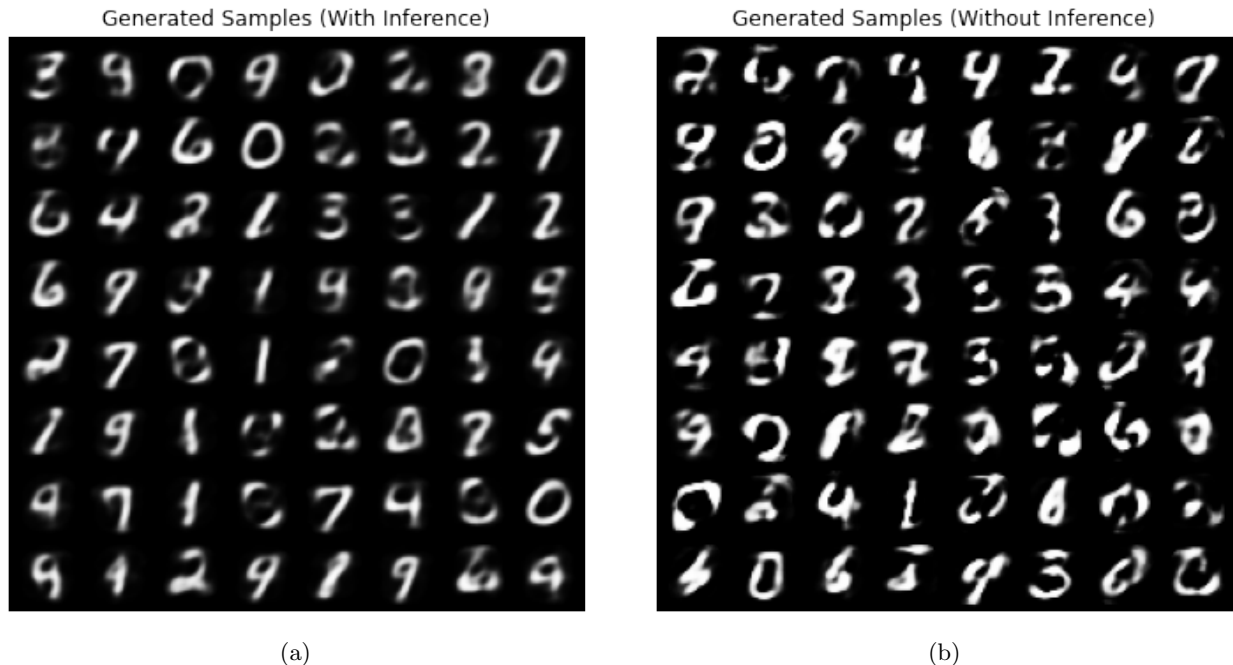


Figure 3: Comparison of results with and without inference.



Figure 4: In-filling missing parts of image.

3.5 Extension to Fashion-MNIST (Original Model)

To assess generalization beyond MNIST, we applied the same model to Fashion-MNIST, a more visually complex dataset containing clothing images (28×28 grayscale).

3.5.1 Setup

- Same architecture: $784 \rightarrow 1000 \rightarrow 100$ encoder, $100 \rightarrow 1000 \rightarrow 784$ decoder
- Trained for 30 epochs, batch size 100, RMSProp ($\text{lr} = 3 \times 10^{-5}$)
- Same inference and Parzen evaluation pipeline

3.5.2 Results on Fashion-MNIST

Epoch	Avg Loss	Parzen LL (With Inference)	Parzen LL (No Inference)
1	564.36	-325.57	-243.15
5	192.18	130.26	209.35
10	82.92	256.91	255.52
15	61.36	271.31	246.19
20	52.91	268.33	238.04
25	48.74	266.28	230.53
30	45.55	258.49	228.53

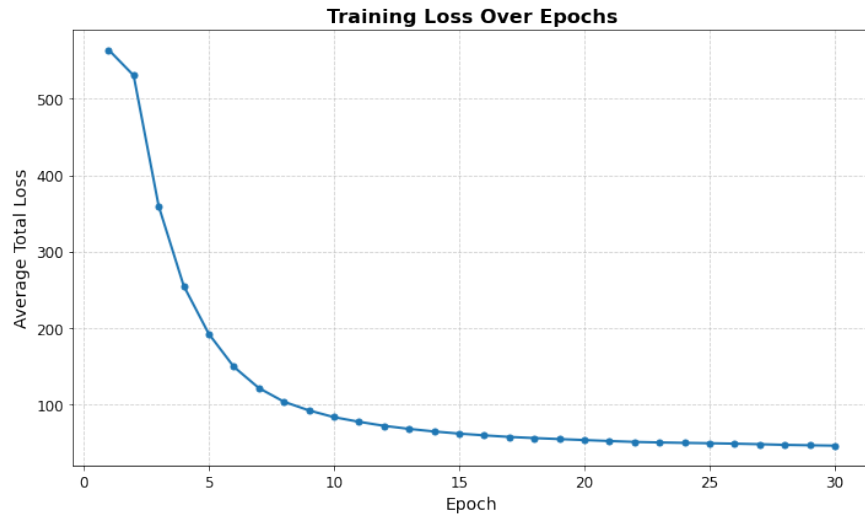


Figure 5: Training Loss over Epochs

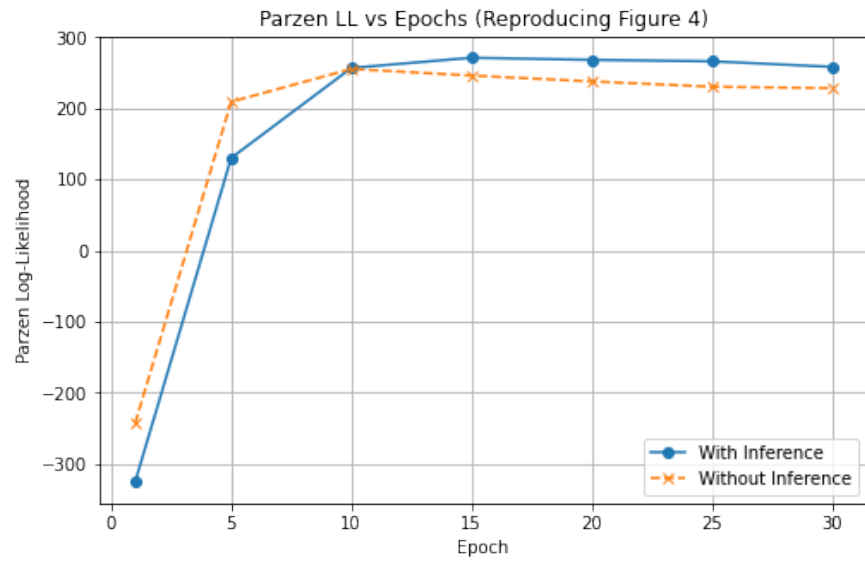


Figure 6: Parzen LL vs Epochs

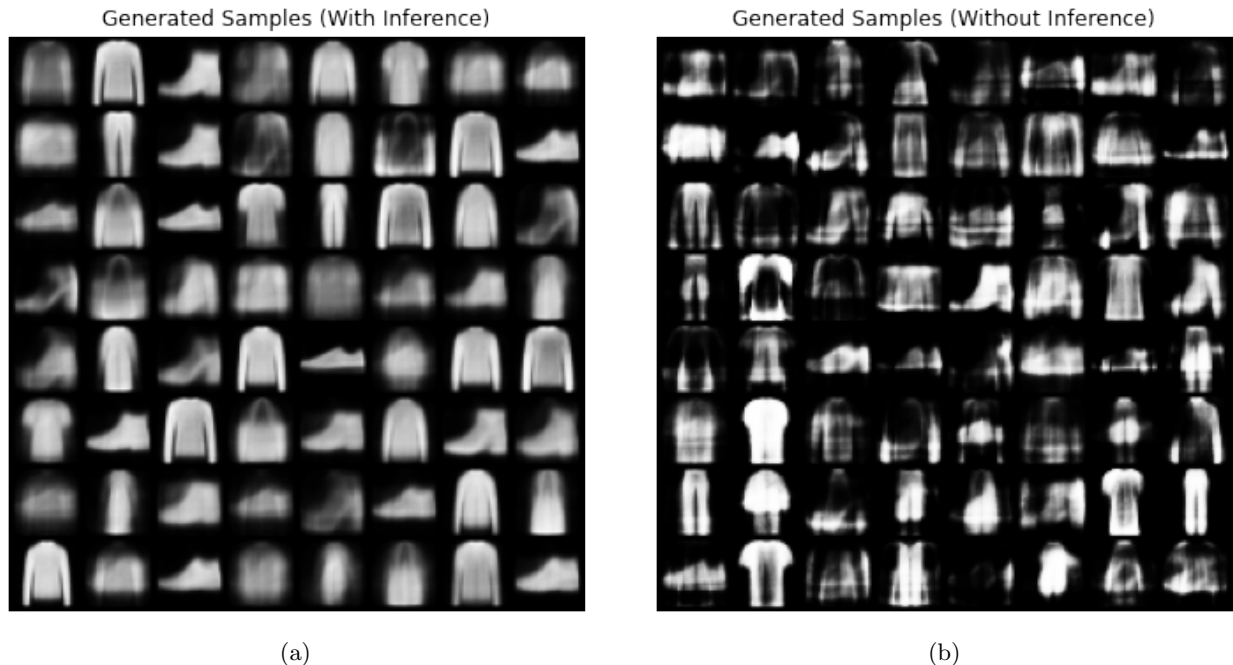


Figure 7: Comparison of results with and without inference.

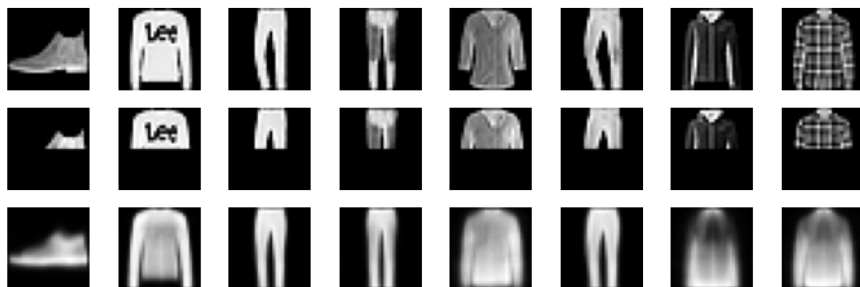


Figure 8: In-filling missing parts of image.

3.5.3 Observations

- The model effectively learned a latent representation for Fashion-MNIST without needing architectural changes.
- Parzen LL peaked between Epochs 15–20, indicating peak generative quality mid-training.
- Inference improved sample quality, although the gain over non-inference generation was smaller than on MNIST—suggesting that more structured datasets benefit more from iterative refinement.

4. Modified Architecture and Comparison

While the original model performed well on both MNIST and Fashion-MNIST, I explored architectural modifications to test how performance scales with increased network depth and capacity, as suggested by the authors.

4.1 Motivation

In the *Joint Denoising Auto-Encoder with Latent Variables* section of the paper, the authors observe that poor sample quality can arise when the prior $p(h)$ generates latent codes not encountered during training. They propose two possible remedies:

1. Increase the capacity of $p(h)$ by using a more expressive or deeper latent model,
2. Improve the inference model $q(h|x)$ by deepening the encoder.

Inspired by this, I introduced a deeper architecture with additional hidden layers in both the encoder and decoder branches, and increased the size of the latent dimensionality.

4.2 Modified Model

Component	Layer Sequence	Activation Functions
Encoder	784 $\rightarrow$ 1000 $\rightarrow$ 500 $\rightarrow$ 150	ReLU, ReLU, Sigmoid
Decoder	150 $\rightarrow$ 500 $\rightarrow$ 1000 $\rightarrow$ 784	ReLU, ReLU, Sigmoid

Table 1: Modified model architecture

Compared to the original model (which used a single hidden layer per path and 100 latent units), this version adds depth and increases the latent dimensionality from 100 to 150. The intent was to give the model more representational power and test whether iterative inference remains beneficial in deeper networks.

4.3 Training on MNIST

The model was trained for 25 epochs (instead of 30) to avoid runtime instability observed beyond epoch 28. Parzen log-likelihood (LL) was evaluated with and without inference across checkpoints.

Epoch	Avg Loss	Parzen LL (With Inference)	Parzen LL (Without Inference)
1	226.20	-172.52	-87.85
5	72.81	27.17	106.78
10	50.29	44.22	136.61
15	45.58	47.10	140.80
20	42.75	42.18	146.63
25	40.10	34.85	144.60

Table 2: MNIST results with modified architecture

Observation: While the modified model started strong, its generative performance (measured via Parzen LL) plateaued early. As training progressed, inference became less effective, and the decoder increasingly learned to map arbitrary latent codes to reasonable outputs. This aligns with the paper’s insight that overly powerful decoders may “bypass” the need for refined latent states, reducing the effectiveness of iterative inference.

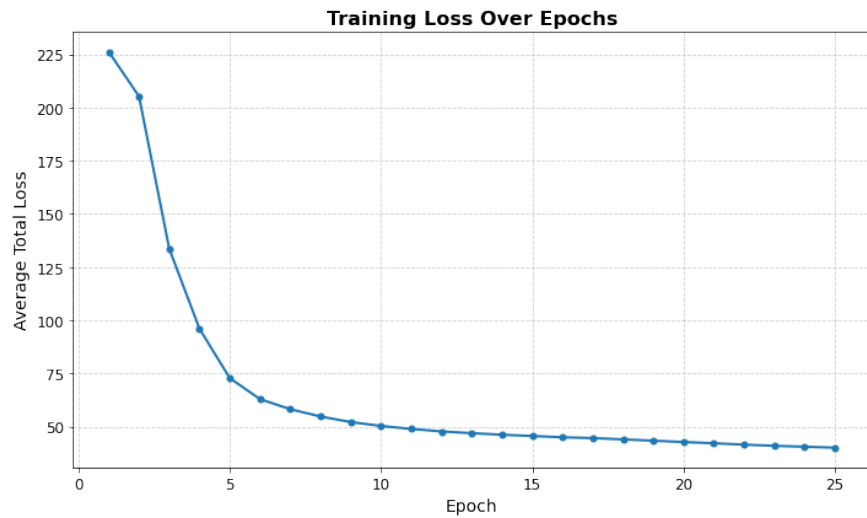


Figure 9: Training Loss over Epochs

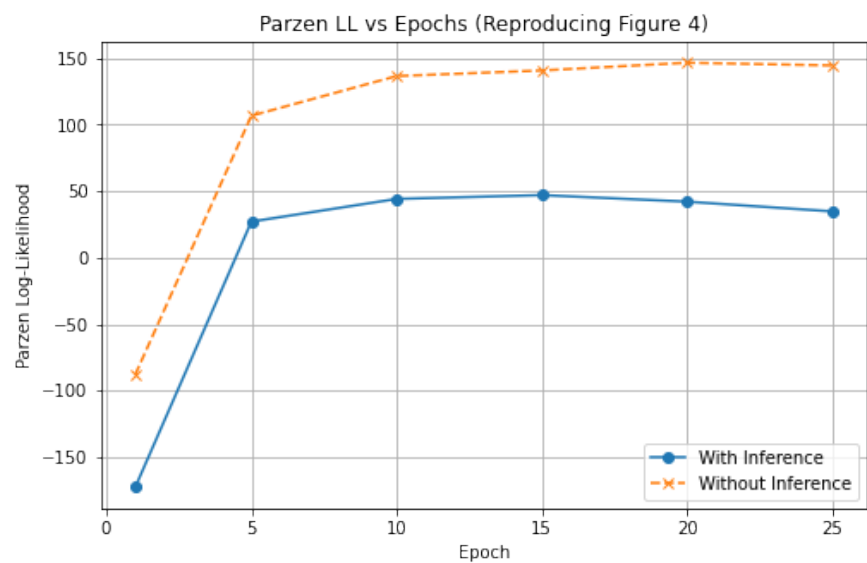


Figure 10: Parzen LL vs Epochs

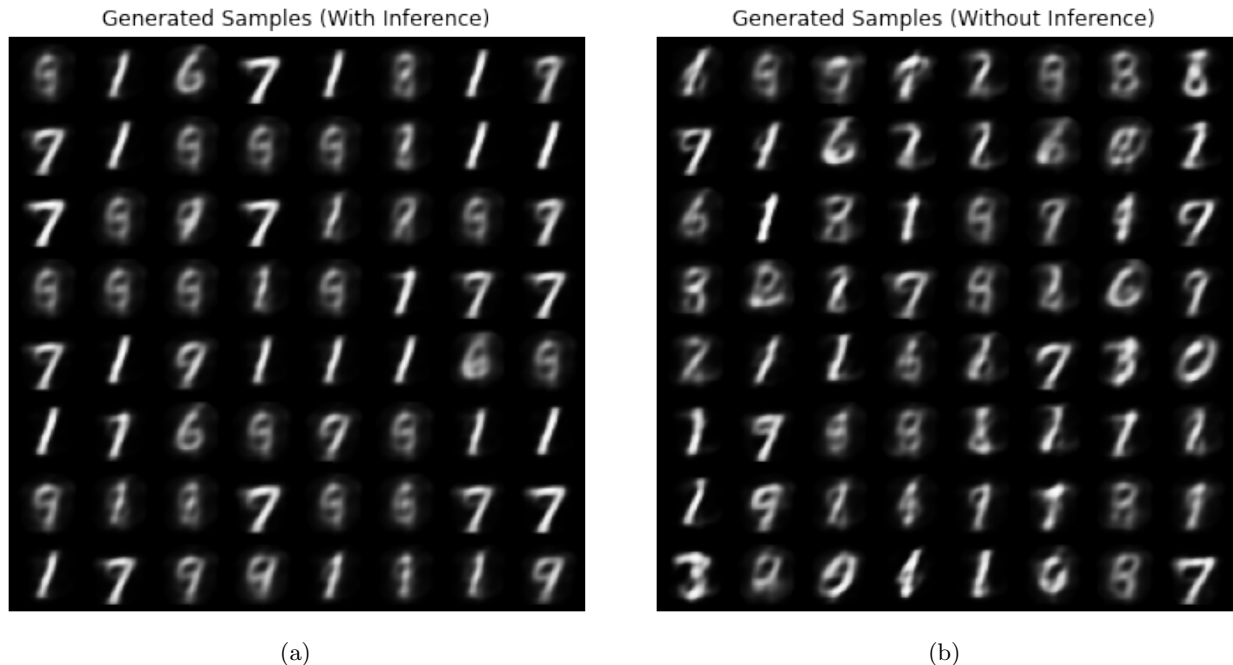


Figure 11: Comparison of results with and without inference.



Figure 12: In-filling missing parts of image.

4.4 Training on Fashion-MNIST

We applied the same modified architecture to Fashion-MNIST, a more complex dataset than MNIST. Training and evaluation procedures remained the same.

Epoch	Avg Loss	Parzen LL (With Inference)	Parzen LL (Without Inference)
1	362.82	-276.26	-50.92
5	47.43	106.39	222.87
10	40.39	183.48	268.61
15	38.85	201.76	279.06
20	39.20	206.25	285.13

Table 3: Fashion-MNIST results with modified architecture

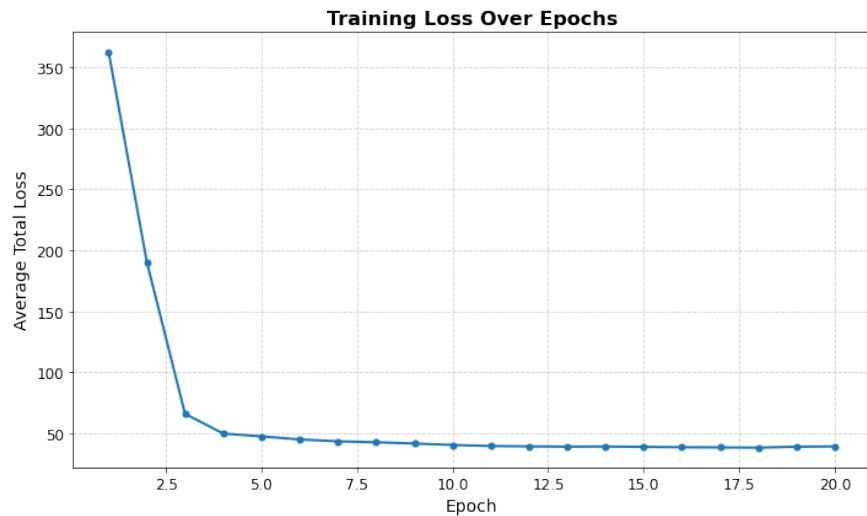


Figure 13: Training Loss over Epochs

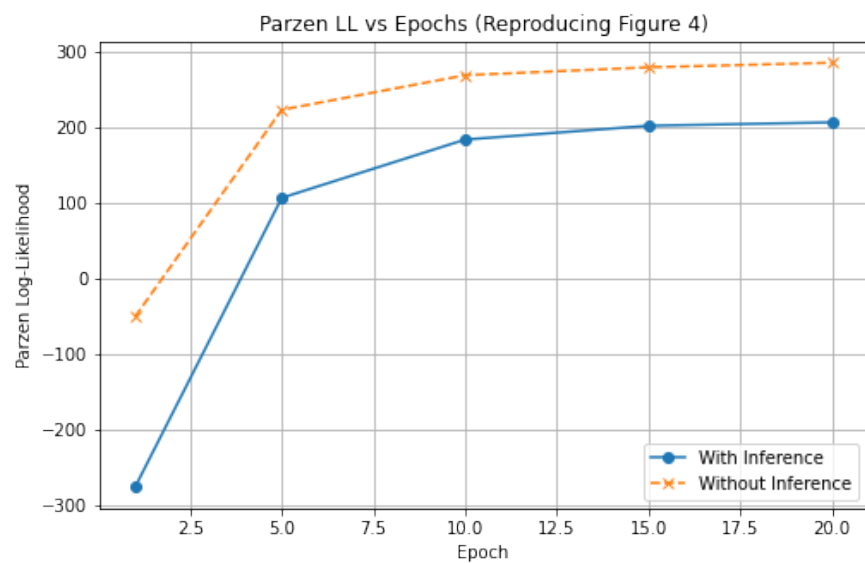


Figure 14: Parzen LL vs Epochs

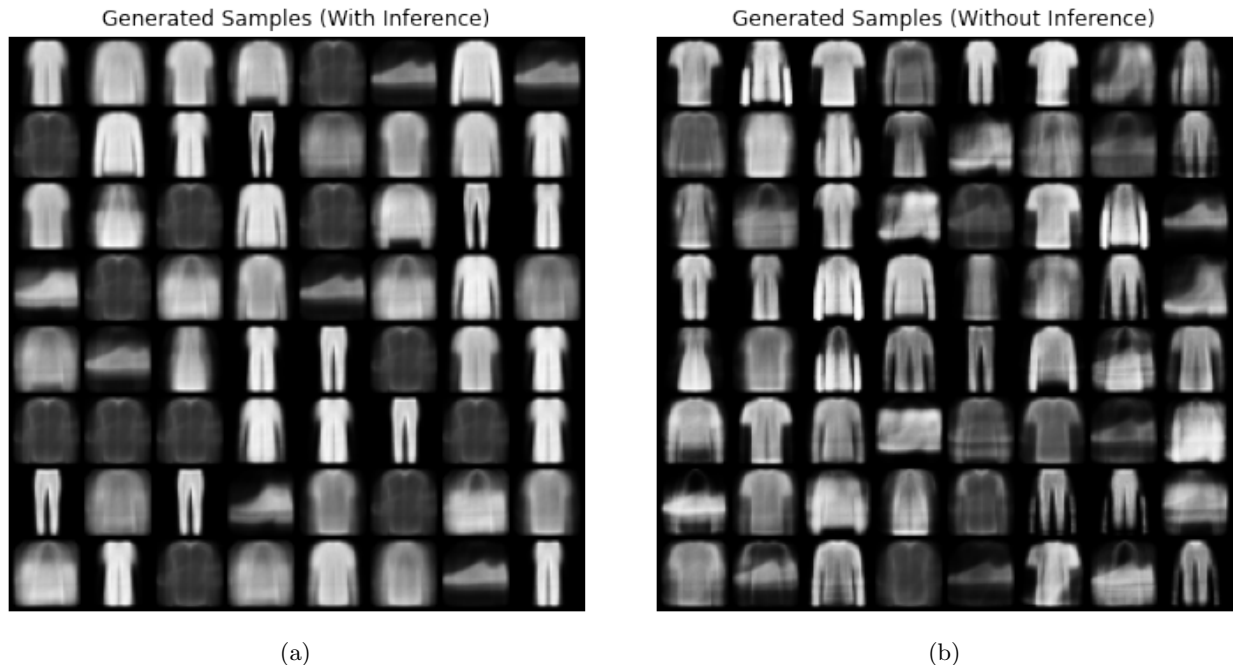


Figure 15: Comparison of results with and without inference.

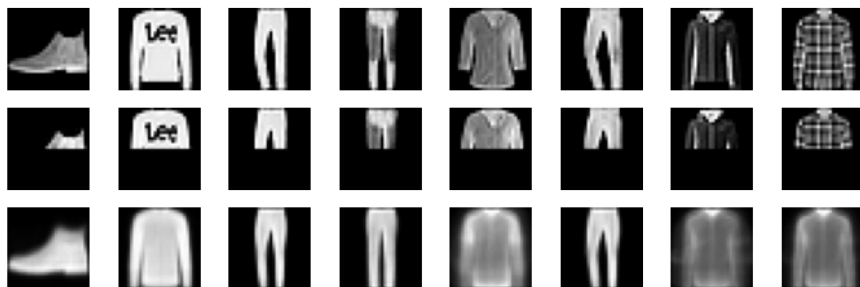


Figure 16: In-filling missing parts of image.

4.5 Insights

- The deeper architecture significantly improved performance on Fashion-MNIST, where higher capacity helped capture more complex patterns.
- However, iterative inference became less effective: the model was able to generate high-quality outputs even without refinement, indicating that the decoder was compensating for inaccuracies in the latent codes.
- This result reinforces the paper’s core idea: inference-based refinement is most useful when the prior $p(h)$ is limited, and the decoder relies more heavily on accurate latent structure.

5. Conclusion and Takeaways

5.1 Conclusion

In this project, I reimplemented the key components of *Towards Biologically Plausible Deep Learning* (Bengio et al., 2015) using PyTorch, starting from scratch due to the original codebase being in Theano. Our

implementation faithfully captured the algorithmic foundations described in the paper:

- Layer-wise denoising autoencoder training for encoder–decoder pairs
- Target propagation as a biologically plausible alternative to backpropagation
- Iterative inference (Algorithm 2) for refining hidden states
- Generative sampling and evaluation using Parzen window log-likelihood

I successfully reproduced the experiments on MNIST, extended them to Fashion-MNIST, and introduced a deeper variant of the model to test the effects of increased representational capacity on sample quality and inference behavior.

5.2 Key Takeaways

- **Target propagation** with denoising objectives is a viable and effective alternative to backpropagation in unsupervised learning.
- **Iterative inference** significantly improves generation quality, particularly when the prior $p(h)$ is weak.
- **Overly expressive decoders** reduce the need for inference and can harm the alignment between $p(h)$ and $q(h|x)$, as reflected by diminished Parzen LL gains.
- The approach **generalizes well to Fashion-MNIST**, demonstrating robustness on more complex image distributions.
- **Parzen window log-likelihood** is an effective tool for evaluating generative models, helping confirm trends described in the paper.

6. References

1. Bengio, Y., Lee, D.-H., Bornschein, J., & Lin, Z. (2015). *Towards Biologically Plausible Deep Learning*. arXiv preprint arXiv:1502.04156.
2. Kingma, D. P., & Welling, M. (2013). *Auto-Encoding Variational Bayes*. arXiv preprint arXiv:1312.6114.
3. Glorot, X., Bordes, A., & Bengio, Y. (2011). *Deep Sparse Rectifier Neural Networks*. Proceedings of the 14th International Conference on Artificial Intelligence and Statistics (AISTATS).
4. Goodfellow, I., Bengio, Y., & Courville, A. (2016). *Deep Learning*. MIT Press. <https://www.deeplearningbook.org>